Energy Efficient Optimal User Association and Node Placement Mechanism for Cooperative UAV Communications

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***Abstract*—To realize energy efficient communication coverage enhancement for cooperative unmanned aerial vehicles (UAVs) communications equipped with multi-beam array antennas, a nested multi-objective mixed integer non-convex model is established to determine the optimal association between user sets and UAV beams as well as the optimal position of UAVs, to minimize the number and total power consumption of UAVs while achieving load balance among UAVs. An alternate joint optimization algorithm that decouples the aforementioned optimization problem to inner and outer layer optimization is proposed. The inner layer alternately optimizes the user association, power consumption and node position of UAVs, while the outer layer optimizes the number of UAVs. Numerical results show that: cooperative UAV communications with multi-beam association and the proposed joint optimization algorithm can flexibly adapt to the demand distribution of complex traffic tasks, reduce total power consumption and achieve load balance among UAVs. These works provide good reference to enhance communication efficiency of cooperative UAV communications.**

***Index Terms*—Cooperative UAV communications, energy efficient resource allocation, multiple beam association.**

I. Introduction

Unmanned aerial vehicles (UAVs), as wireless communication networks of air base stations, can dynamically and selectively increase the network capacity due to their characteristics of rapid deployment, flexible relocation, good line-of-sight propagation path and more. This has been considered as one of basic capabilities of the 5G/6G cellular network [1], [2], [3], [4], [5], [6]. The massive multiple-input multiple-output (MIMO) technology, also as one of key technologies for future networks [7], [8], [9], [10], [11], [12], is considered as the key technology to expand the capacity of the system and improve spectrum utilization. By combining the massive MIMO technology with cooperative UAV communication, UAVs can not only find a good channel environment with flexible maneuverability, but also improve signal power gain and suppress interference by specifying beam direction. Combining the capacity gain of the massive MIMO technology with the mobility of UAVs can provide on-demand communication services for users on the ground accurately and efficiently [13], [14], which is of great practical significance in emergency communication or other fields [15], [16], [17], [18], [19], [20], [21]. Deploying UAV base stations can not only meet the sudden high demand for mobile data and expand the capacity of cellular networks but also enable timely recovery after the event ends. UAV-assisted wireless communication can be roughly divided into three categories based on the role of UAVs: deployed in areas with existing communication infrastructure to assist air base stations; when the communication infrastructure is partially or completely damaged, deploying drone air base stations can quickly restore communication services or alleviate the high load pressure on ground base stations in extremely crowded areas; deployed between two or more users to serve as an auxiliary relay, and when two or more users are far apart or there are obstacles preventing signal propagation, relay UAVs can be deployed to reduce communication interruption rates; and assist in information dissemination and data collection. In sensor networks, UAVs can efficiently collect data from ground sensor nodes and send it to designated nodes with their flexible mobility [22], [23], [24].

To fully utilize the advantages of the UAV cooperative communication after the introduction of multi-beam array antennas, this paper focuses on jointly planning and deployment problems of the multi-beam UAV cooperative communication network, such as user-beam association, total network power consumption, the number of UAVs and position optimization. As for the

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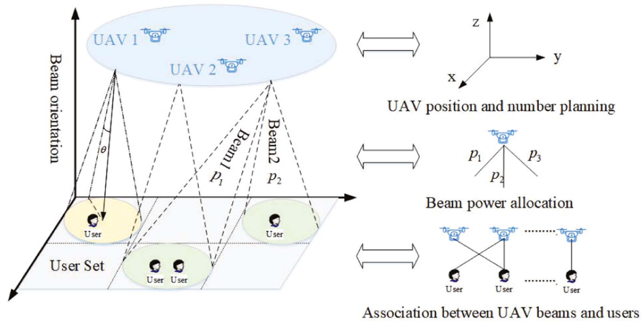

Fig. 1. Multi-beam UAV collaboration provides on-demand communication services for users on the ground.

user association problem, references [25] and [26] studied the user association problem from the perspective of minimizing the communication rate of users, while references [27] and [28] considered the user association problem from the perspective of load balance. [24] proposed a distributed algorithm to jointly optimize the spatial position and user association of UAVs, thereby maximizing the total network rate. [29] and [30] respectively used Neural Network (NN)-based and Deep Learning (RL)-based user association algorithm to solve user association problem. As for the power allocation problem, references [31], [32], [33] proposed an idea of converting the power allocation problem of non-convex optimization into convex optimization, including Lagrange duality theory and successive convex approximation. [34] studied the joint resource allocation problem of UAVs with synchronous wireless information and power transmission in multicast scenarios. As for the UAV position planning, reference [35] proposed a position deployment scheme for minimizing coverage. [36] optimized the deployment of UAV positions to achieve maximum coverage of users with minimal transmission power. [37] studied how to effectively and efficiently deploy UAVs for load balancing. Reference [38] maximized the total downlink rate by optimization of UAV positions and available resources. However, the optimization of the number of UAVs is less studied. The most common method is to estimate the number of required UAVs according to the number of users and the user demands. [39] studied the problem of base station planning in cooperative communication with the goal of minimizing total network power consumption. In addition to studying the location planning of base stations, the study also provided a method for optimizing the number of base stations. In the above studies, only the single association (that is, one user can only be associated with one base station) is taken into consideration, but the multi-association (that is, one user can be associated with multiple serving base stations) introduced by the multi-beam UAV cooperative communication, is not taken consideration. Consequently, it is impossible to completely model the system performance gain caused by the flexibility of multi-association [40].

In view of the above demands and challenges, the contributions of the paper can be summarized as follows:

- Considering the multi-association between users and UAVs, this paper focuses on the optimal resource allocation problem of the multi-beam UAV cooperative communication network.
- To minimize the number of UAVs and the total network power consumption and achieving load balance among UAVs while meeting the users' communication demands by jointly optimizing the user association, power allocation and the position and number of UAVs, an alternate optimization scheme for decoupling the optimization problem to inner and outer layer optimization is proposed.
- Based on our simulation results, the proposed multi-beam UAV cooperation and joint optimization scheme can flexibly adapt to the distribution of complex traffic distributions, reduce the power consumption and achieve the load balance among UAVs.

The rest of the paper is organized as follows. In Section II, the system model and the problem model for the optimal resource allocation are introduced. In Section III, an alternate joint optimization algorithm is proposed. Numerical results are provided and analyzed in Section IV. Finally, Section V concludes the paper.

II. System Model and Probelm Modeling

A. System Model

Fig. 1 shows a network model in which the multi-beam UAV cooperation provides temporary communication services for users on the ground on demand. M UAVs are included and represented by $I = \{i_1, i_2, i_3, \ldots, i_M\}$. Each UAV has N beams whose orientation can be adjusted freely, and these beams are represented by $J = \{j_1, j_2, j_3, \ldots, j_N\}$. User regions on the ground are divided into grids according to the coverage of UAV beams, and the users belonging to the same grid constitute a user set which is represented by $K = \{k_1, k_2, k_3, \ldots, k_s\}$. The number of users in each grid region is counted and represented by $U = \{u_1, u_2, u_3, \ldots, u_s\}$. A 3D Cartesian coordinate system is established by using the boundaries of the rectangular region where the user is located by x-axis and y-axis and using the direction where the UAV i is located as z-axis. The position coordinates of UAV base stations and users at the grid point are represented by $[x_i, y_i, H]$ and $[x_k, y_k, 0]$, respectively. The position of the grid point is represented by the center point of each divided region.

The system model obtained after grid division uses the user sets in the grids as association objects. Compared with the existing system models using single user as association object, this system model has the following advantages.

1) The movement of users in a small range is supported. The number of users in the grid will remain unchanged as long as the movement range of the users is not beyond this grid, avoiding network deployment readjustment resulted from the user's frequent joining in or exiting from the network node when the user is moving.
2) The beam orientation problem is simplified. When considering the direction of the beam, studying the correlation problem from the perspective of a single user can become

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TABLE I
SIMULATION PARAMETERS

Parameter name	Parameter symbol	Parameter value
Signal-to-noise ratio	β/δ^2	50(dB)
Loss coefficient	α	2
Maximum transmission power for UAV	p_{max}	1(W)
The number of users	U	100
The number of user sets	S	9
UAV height	H	100(m)

very complex and difficult to solve. But in this model, each beam exactly covers one network, so the UAV beam orientation problem can be converted into an association problem between UAV beams and user sets, i.e., a user association.

3) The multi-association modeling approach is simplified. When studying multi-association problems (i.e. allowing a user to associate multiple access points simultaneously), this model understands the single association problem as a grid point can only be covered by one beam, while multi-association allows a grid point to be covered by multiple beams simultaneously.
4) Interference modeling is simplified. Since a single beam covers only one user set, in the consideration of the interference power, it is different from before that all power and Gaussian white noise from its own unrelated access points need to be considered as interference power, and only Gaussian white noise needs to be considered, but the interference from other signals is ignored. Accordingly, the formulation and problem solution are simplified.

To sum up, the proposed system model can support the studies on the resource allocation problem of the multi-beam UAV cooperative communication and gain advantages by simplifying problem modeling and solution.

B. Modeling Assumption

Next, the resource optimization model for the multi-beam UAV cooperative communication network will be established based on the system model. The modeling assumptions used here are given below, including:

1) the position distribution of users on the ground is fixed and known;
2) the UAV has a fixed number of beams, and the adjustment of beam orientation is discretized, that is, the beams point from one grid to another grid without overlaps;
3) each beam of the UAV can cover only one grid, but one grid can be covered by multiple beams, thus realizing multi-association;
4) when a grid is covered by multiple beams, the load is evenly distributed to the beams;
5) the number of users included in each grid indicates the communication demand of the current grid;
6) the path loss propagation model employs a free space propagation model.
7) signal interference between grids is not considered.

Table I shows the main parameters used and their meanings.

C. Nested Multi-Objective Mixed Integer Non-Convex Optimization Model

A nested multi-objective mixed integer non-convex optimization model that considers both the position and number of UAVs, power allocation, and the association between UAV beams and users have been established. This model aims to achieve load balance among UAVs, minimize the total network power consumption, and reduce the number of UAVs while meeting users' communication demands, as shown in Formulas (1)–(8) below.

$$\min_{\rho,p,[x,y]} M \tag{1}$$

$$\min_{\rho,p,[x,y]} \sum_{i=1}^{M}\sum_{j=1}^{N} p_{ij} \tag{2}$$

$$\min_{\rho,p,[x,y]} \frac{1}{M}\sum_{i=1}^{M}\left(\sum_{j=1}^{N}\sum_{k=1}^{S}\rho_{ijk}\overline{u_k} - \frac{1}{M}\sum_{i=1}^{M}\sum_{j=1}^{N}\sum_{k=1}^{S}\rho_{ijk}\overline{u_k}\right)^2 \tag{3}$$

$$s.t. \sum_{i=1}^{M}\sum_{j=1}^{N}\rho_{ijk}\log_2\left(1+\frac{p_{ij}g_{ik}}{\delta^2}\right) \geq u_k R_{th}, \forall k \tag{4}$$

$$\sum_{k=1}^{S}\rho_{ijk} \leq 1, \forall i,j \tag{5}$$

$$\overline{u_k} = u_k / \sum_{i=1}^{M}\sum_{j=1}^{N}\rho_{ijk}, \forall k \tag{6}$$

$$\sum_{j=1}^{N} p_{ij} \leq p_{\max}, \forall i \tag{7}$$

$$\rho_{ijk} \in \{0,1\}, \forall i,j,k \tag{8}$$

The meanings of the above formula are summarized below.

1) The Formula (1) is a first objective function which requires the minimum number of UAVs.
2) The Formula (2) is a second objective function which requires the minimum total network power consumption.
3) The Formula (3) is a third objective function which requires the minimum load difference (i.e., load balance) among UAV base stations, the load of UAVs being represented by the number of users associated with UAVs.
4) The Formula (4) is a first constraint indicating that the actual communication rate of each grid must be greater than or equal to the communication rate demand threshold. Here, each beam is assumed as unit bandwidth, and δ^2 represents the Gaussian white noise. The g_{ik} is calculated by the following formula:

$$g_{ik} = \frac{\beta}{(d_{ik})^\alpha} = \frac{\beta}{[(x_i-x_k)^2+(y_i-y_k)^2+H^2]^{\alpha/2}}. \tag{9}$$

where β represents a channel power gain value at the specified reference distance of 1m, α represents a path loss coefficient, and d_{lk} represents the distance between the UAV i and the user set j.

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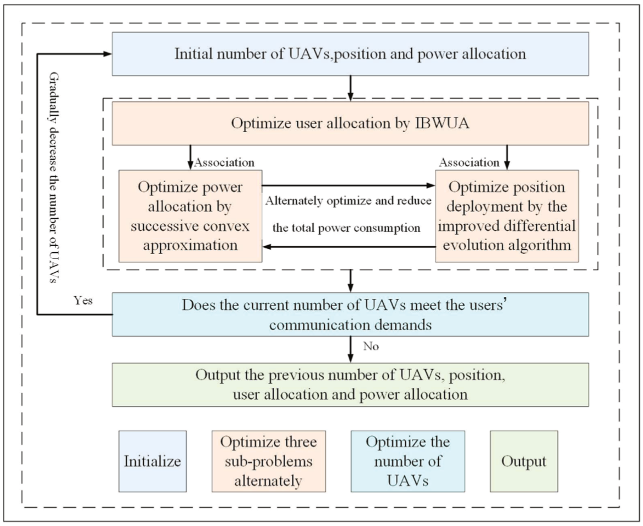

Fig. 2. Flow chart of joint optimization scheme.

5) The Formula (5) is the second constraint indicating that each beam can cover only one network.
6) The Formula (6) indicates that the load is evenly distributed to multiple beams when the user set is covered by the multiple beams.
7) The Formula (7) is a third constraint which is a power constraint indicating that the maximum value of the total power that can be allocated to the beams by each UAV is $p_{\max}$.
8) The Formula (8) is a fourth constraint indicating that the association variable can only be 0 or 1.

It is noted that the constraint is not included here.

$$\sum_{i=1}^{M}\sum_{j=1}^{N}\rho_{ijk}=1. \tag{10}$$

This constraint means that each grid can be covered by only one beam. The absence of this constraint in this model allows one grid point to be covered by multiple beams, thus realizing multi-association.

III. Alternate Joint Optimization Algorithm

Since multiple variables in the nested multi-objective mixed integer non-convex optimization model are coupled together and difficult to solve by standard convex optimization, a joint optimization algorithm based on alternate optimization is proposed, as shown in Fig. 2. This algorithm decomposes the original problem into multiple sub-problems, including the beam and user set association problem, the power allocation problem, and the UAV position and quantity planning problem. Then, each remaining sub-problem is solved while the other sub-problems are fixed, and finally, alternating optimization is continuously carried out. The joint optimization algorithm decouples the nested optimization model to inner and outer layer optimization. The inner layer optimization fixes two sets among three sets of variables ρ, ρ and UAV position $[x, y, H]$ to iteratively solve the optimal values of the remaining set of variables. The outer layer optimization decreases the number of UAVs based on the result of inner layer alternate optimization until the minimum number of UAVs meeting the users' communication demands is obtained.

A. Sub-Problem 1: Optimal User Association

The model and solution method for the user association problem will be given. After the position and number of UAVs and the power allocation are fixed, the association problem between beams and user sets is simplified to achieve load balance among UAVs under the condition of meeting the users' communication rate demands:

$$\begin{aligned}
&\min_{\rho}\frac{1}{M}\sum_{i=1}^{M}\left(\sum_{j=1}^{N}\sum_{k=1}^{S}\rho_{ijk}\overline{u_k}-\frac{1}{M}\sum_{i=1}^{M}\sum_{j=1}^{N}\sum_{k=1}^{S}\rho_{ijk}\overline{u_k}\right)^2\\
&s.t. \sum_{i=1}^{M}\sum_{j=1}^{N}\rho_{ijk}\log_2\left(1+\frac{p_{ij}g_{ik}}{\delta^2}\right)\geq u_k R_{th}, \forall k\\
&\sum_{k=1}^{S}\rho_{ijk}\leq 1, \forall i, j\\
&\overline{u_k}=u_k/\sum_{i=1}^{M}\sum_{j=1}^{N}\rho_{ijk}, \forall k\\
&\rho_{ijk}\in\{0,1\}, \forall i, j, k
\end{aligned} \tag{11}$$

This problem is a typical NP problem. A meta-heuristic algorithm is very effective in solving this problem compared with conventional algorithms [41], so a near-optimal solution can be obtained in polynomial time instead of exponential time. This problem will be transformed before solution, and the constraints are converted into an objective function by a penalty method. Thus, this problem is equivalent to:

$$\begin{aligned}
&\min_{\rho} f(\rho)+\sum_{k=1}^{S}\eta_k[\min(0, h_k(\rho))]^2\\
&s.t. \sum_{k=1}^{S}\rho_{ijk}\leq 1, \forall i, j\\
&\rho_{ijk}\in\{0,1\}, \forall i, j, k
\end{aligned} \tag{12}$$

where

$$\begin{aligned}
f(\rho)=\frac{1}{M}\sum_{i=1}^{M}\Bigg(&\sum_{j=1}^{N}\sum_{k=1}^{S}\rho_{ijk}(u_k/\sum_{i=1}^{M}\sum_{j=1}^{N}\rho_{ijk})\\
&-\frac{1}{M}\sum_{i=1}^{M}\sum_{j=1}^{N}\sum_{k=1}^{S}\rho_{ijk}\left(u_k/\sum_{i=1}^{M}\sum_{j=1}^{N}\rho_{ijk}\right)\Bigg)^2
\end{aligned} \tag{13}$$

$$h_k(\rho)=\left(\sum_{i=1}^{M}\sum_{j=1}^{N}\rho_{ijk}\log_2\left(1+\frac{p_{ij}g_{ik}}{\delta^2}\right)\right)-u_k R_{th} \tag{14}$$

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Algorithm 1: IBWA Algorithm.

1: initialize the whale position information X_i^0, $i = 1, 2, 3, \ldots, num$, set the maximum number of iterations $Iters$, and assume $l = 1$
2: while $l < Iters$ do
3: map the position information to an association variable matrix, and calculate the corresponding objective function value
4: select the head whale X_{best}^l
5: update all whale positions according to the rules for the whale optimization algorithm
6: calculate the updated objective function value of each whale individual
7: adjust the population number according to the population adaption rule
8: assume $l = l + 1$
9: end while
10: output the association between UAV beams and user sets

In the formula, $\eta_k \geq 0$ is a penalty factor. When $h_k(\rho) < 0$, a large "penalty" is applied to the augmented objective function, so this solution will not be selected. Of course, it is noted that the penalty factor should not be too large, or the optimality of the obtained solution will be affected.

The whale optimization algorithm (WOA) is an algorithm proposed to mimic the behavior of humpback whales in hunting prey. In the algorithm, the position information of each humpback whale can be regarded as a feasible solution. During the hunting process, each whale can choose between two behaviors: moving towards other whales to surround the prey or generating bubbles through circular swimming to drive away prey; the whale that chooses the first behavior to surround its prey can either approach the whale in the current best position or a random whale, and this choice is random. Due to this unique mechanism for obtaining near-optimal solutions, the WOA provides good global search capability. The problem (13) is solved using an improved WOA, named Improved Binary WOA for User Association (IBWA). First, the mapping of the problem to the whale foraging model is completed: the correlation between the UAV beam and the user set corresponds to an individual whale, and the whale foraging process represents the search for the optimal solution. During the search process, the location information of each whale corresponds to a feasible solution for the correlation relationship in the optimization process, and the location information of the head whale represents the best solution for the user correlation relationship in the current iteration. The objective function value of the head whale is the minimum load balancing value among UAVs in the current iteration process. The IBWA algorithm is with the following main improvements:

1) The constraints (5) and (8) are strong constraints, so it is difficult to find feasible solutions by random searching. According to the constraint (8), the successive whale position information is mapped to an association variable matrix $\rho_{MN,S}$:

$$\max\{\rho_{i,1}, \rho_{i,2}, \rho_{i,3}, \ldots, \rho_{i,S}\} = 1, \forall i = 1, 2, \ldots, MN \tag{15}$$

In other words, the largest value in each row of the matrix is 1, while other values are automatically set as 0. Thus, a binary association variable matrix is obtained for calculating the objective function value. Additionally, according to the constraint (5), the variable with a value of 1 is changed to 0 at a certain probability:

$$P = \frac{0.5}{\text{The number of iterations}} \tag{16}$$

The probability will gradually decrease from 0.5 with the increase of the number of iterations.

2) The scale of the whale population is changed adaptively, and the probability of global optimization is increased by adaptively changing the size of the search space. The basic idea of the population adaption strategy lies in that [42]: if the position information of the head whale is updated for two consecutive iterations and the current population number is greater than the predetermined minimum population number, the number of whales in the whale population deceases; and, if the position information of the head whale is not updated in any iteration and the current population number is less than the predetermined maximum population number, the number of whales in the whale population increases.

The IBWA algorithm process is as follows: after initializing the whale position information, the whale position information is transformed into the association between UAV beams and user sets; calculate the fitness of each whale and select the head whale; update the position of the whale in the next iteration process using the location information of the head whale; cycle the above process until the final iteration; translate the final whale position information into the correlation between the drone beam and the user set and output it. The specific process of the IBWA algorithm is as shown in Algorithm 1.

B. Sub-Problem 2: Optimal Power Allocation

The model and solution method for the power allocation problem will be given. The position and number of UAVs and the association between UAV beams and user sets are fixed, and the UAV power allocation problem is simplified to minimize the total power consumption of UAVs under the condition of meeting the users' communication rate demands:

$$\begin{aligned} &\min_{p} \sum_{i=1}^{M}\sum_{j=1}^{N} p_{ij} \\ &s.t. \sum_{i=1}^{M}\sum_{j=1}^{N} \rho_{ijk} \log_2\left(1 + \frac{p_{ij} g_{ik}}{\delta^2}\right) \geq u_k R_{th}, \forall k \\ &\sum_{j=1}^{N} p_{ij} \leq p_{\max}, \forall i \end{aligned} \tag{17}$$

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Since the logarithmic function of the variable Pij in the first constraint is a concave function, this problem is obviously not a convex optimization problem. So, it is impossible to directly solve this problem by using the typical convex optimization theory. Assuming that:

$$h(p_{ij}) = u_k R_{th} - \sum_{i=1}^{M}\sum_{j=1}^{N} \rho_{ijk} \log_2\left(1 + \frac{p_{ij} g_{ik}}{\delta^2}\right). \quad (18)$$

If p_{ij}^l is defined as the power value of each beam allocated by each UAV after l iterations and Taylor expansion is performed on $h(p_{ij})$ at p_{ij}^l, the following inequality is tenable:

$$h(p_{ij}) \leq h(p_{ij}^l) + h'(p_{ij}^l)(p_{ij} - p_{ij}^l). \quad (19)$$

Hence, the problem

$$\begin{aligned} &\min_{p} \ \sum_{i=1}^{M}\sum_{j=1}^{N} p_{ij} \\ &s.t. \ \ h^{ub}(p_{ij}), \forall k \\ &\sum_{j=1}^{N} p_{ij} \leq p_{\max}, \forall i \end{aligned} \quad (20)$$

is the upper bound of the problem (17), where:

$$\begin{aligned} h^{ub}(p_{ij}) = u_k R_{th} - \sum_{i=1}^{M}\sum_{j=1}^{N} \rho_{ijk} \log_2\left(1 + \frac{p_{ij}^l g_{ik}}{\delta^2}\right) \\ + \sum_{i=1}^{M}\sum_{j=1}^{N} \rho_{ijk} \frac{g_{ik}}{\delta^2 + g_{ik} p_{ij}^l}(p_{ij} - p_{ij}^l). \end{aligned} \quad (21)$$

The problem (20) is a typical convex optimization problem because the objective functions and the constraints are all linear functions of the association variable P_{ij}. Although this convex optimization problem can be solved by many existing typical algorithms, this problem is solved by an interior point method here. However, the problem (20) is an approximate transformation of the problem (17), which is to be approximately solved by successive convex approximation to obtain the local optimal solution of the original problem. The power allocation algorithm process is as follows:

1) Perform first-order Taylor expansion on the first constraint to obtain the global upper bound function of the original concave function, and construct an approximate convex function problem (20).
2) Initialize the initial power allocation value p^l that meets the constraints, set the minimum reduction value ε, and set $l = 1$.
3) Take p^l as the initial point of the problem and use the interior point method to solve the convex optimization problem (20) to obtain the optimal power allocation value p^*.
4) If the result is less than the minimum reduction value, stop the loop. Otherwise, update the power allocation value $p^{l+1} = p^*$ and let $l = l + 1$, repeat steps 2 to 3.
5) The output result is the optimal solution for the power allocation of each beam by the UAV.

Algorithm 2: Power Allocation Algorithm.

1: initialize the initial power value p^l meeting the constraint, assume $l = 1$, and set the minimum reduction value ε
2: while the reduction of the total power consumption is less than ε do
3: solve the problem (20) by the interior point method to obtain an optimal power allocation value p^*
4: update the initial value $p^{l+1} = p^*$ of the next power allocation
5: assume $l = l + 1$
6: end while
7: output p^{l+1} as the optimal power allocation of the problem (21)

The power allocation algorithm based on successive convex approximation is shown in Algorithm 2.

C. Sub-Problem 3: Optimal Position Deployment

The model and solution method for the position deployment problem will be given. The number of UAVs, the power allocation and the association between UAV beams and user sets are fixed, and the optimal deployment problem of UAV positions is simplified as:

$$\begin{aligned} &\min_{[x,y]} \sum_{i=1}^{M}\sum_{j=1}^{N} p_{ij} \\ &\min_{[x,y]} \frac{1}{M} \sum_{i=1}^{M} \left(\sum_{j=1}^{N}\sum_{k=1}^{S} \rho_{ijk}\overline{u_k} - \frac{1}{M}\sum_{i=1}^{M}\sum_{j=1}^{N}\sum_{k=1}^{S} \rho_{ijk}\overline{u_k} \right)^2 \\ &s.t. \sum_{i=1}^{M}\sum_{j=1}^{N} \rho_{ijk} \log_2(1 + f(x,y)) \geq u_k R_{th}, \forall k \end{aligned} \quad (22)$$

where

$$f(x,y) = \frac{\beta p_{ij}}{\delta^2[(x_i - x_k)^2 + (y_i - y_k)^2 + H^2]^{\alpha/2}}. \quad (23)$$

Since the variables ρ and p have been determined, the feasible solution is only required to meet the constraints. With reference to the use of the penalty function in Section III-A, this problem can be converted into:

$$\max_{[x,y]} \sum_{k \in S} [\varphi(g_k(x,y)) g_k(x,y)] \quad (24)$$

where

$$g_k(x,y) = \sum_{i=1}^{M}\sum_{j=1}^{N} p_{ijk}\omega_{ij} \log_2(1 + f(x,y)) - u_k R_{th} \quad (25)$$

$$\varphi(g_k(x,y)) = \begin{cases} 1, \text{if } g_k(x,y) \geq 0 \\ \nu, \text{if } g_k(x,y) < 0 \end{cases} \quad (26)$$

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The ν is a penalty factor which is a sufficiently large positive number. Some changes have been made to the standard penalty function, and such changes are related to the whole problem. When the association between UAV beams and grid points and the bandwidth allocation of beams have been determined, the UAV position optimization sub-problem is required to meet only the constraint (4), but more than one feasible solution meets this requirement. The final goal of the problem is to minimize the total network power consumption, so it is necessary to find, from all the feasible solutions, a solution that increases the communication rate of users as much as possible. Thus, the users' demands for the same communication rate can be met at less power consumption under the same UAV placement, so that the total power consumption can be minimized during joint optimization.

The differential evolution algorithm, as a global optimization algorithm based on population adaption, has the advantages of fewer parameters and higher optimization capability in comparison to the genetic algorithm and the ant colony algorithm. The optimal position deployment problem of UAVs will be solved by the differential evolution algorithm here. The steps of the improved differential evolution algorithm are shown in Algorithm 3.

The differential evolution algorithm process is as follows:

1) Population initialization: Considering that the initial information has a great impact on the probability of obtaining the global optimal solution by this algorithm, the mechanism of randomly generating initial information by the differential evolution algorithm is improved such that the initial UAV position information is generated based on distance clustering through a k-means algorithm by fully utilizing the known information (i.e., the user position distribution).
2) Perform population variation: Randomly select 3 individuals from the population, and use these three individuals to generate variation vectors according to differential evolution rules.
3) Perform population crossover based on differential evolution rules.
4) After completing the crossover, select individuals from the next generation population based on the greedy policy. In the iterative process, it is required that the next generation must be better or equal to the previous generation, and through repeated variation, crossover, and selection, the population ultimately reaches its optimal state.

Algorithm 3: UAV Position Optimization Algorithm.

```
1: set the maximum number of iterations G, and perform
   population initialization by the k-means algorithm
2: for ge = 1 : G do
3:    perform population variation and crossover, obtain
      the new population M_m(ge)
4:    calculate the objective function value corresponding
      to each individual
5:    select individuals by a greedy policy to obtain a new
      population
6: end for
7: obtain the final population D, and select individuals
   with the smallest objective function value from the
   final population as results
8: output the position coordinates of edge nodes
```

Algorithm 4: Alternate Optimization Algorithm.

```
1: initialize the number of UAVs, UAVs, user position and
   beam power
2: do {
3:   fix the power allocation p and UAV position [x, y], and
     calculate the association according to the Algorithm 1
4:   calculate the total power consumption at this time
5:   do {
6:     fix the power allocation and user association, and
       optimize the UAV position according to the
       Algorithm 3
7:     fix the user association and UAV position, and
       optimize the resource allocation according to the
       Algorithm 2
8:     calculate the total power consumption at this time
       and obtain the reduction of the total power
       consumption
9:     }while(the reduction of the total power consumption
       is greater than the predetermined threshold)
10:  calculate the actual communication rage of each user
     set at this time
11: }while(the communication demand of each user is met)
12: output the number of UAVs, association variable, power
    allocation and UAV position
```

D. Iteration Process of Alternate Joint Optimization

So far, the methods for solving three sub-problems have been given. In terms of the optimal or minimal number of UAVs, it is obtained by gradually decreasing the number of UAVs according to the results of inner layer alternate optimization, that is, when optimized, as long as the present number of UAVs is enough to meet the communication demands, the number of UAVs can be decreased, as shown in red part of Fig. 2. Algorithm 4 shows the proposed complete alternate joint optimization algorithm. The total network power consumption is reduced by alternately optimizing the power allocation and position deployment in the process of decreasing the number of UAVs until it converges. It is noted that: since the solution obtained by the alternate optimization algorithm is a near-optimal solution, the optimal performance will be affected by the setting of initial variables, such as initial positions of UAVs.

IV. Numerical Analysis

The numerical simulation results and the analysis of the simulation results will be mainly described. The simulation scenario is a 100×100 m ground region. Multiple UAVs with a fixed number of beams hover in the air at 100 m above this region. The downlink cellular network composed of these UAVs provides temporary communication services for the users on the ground.

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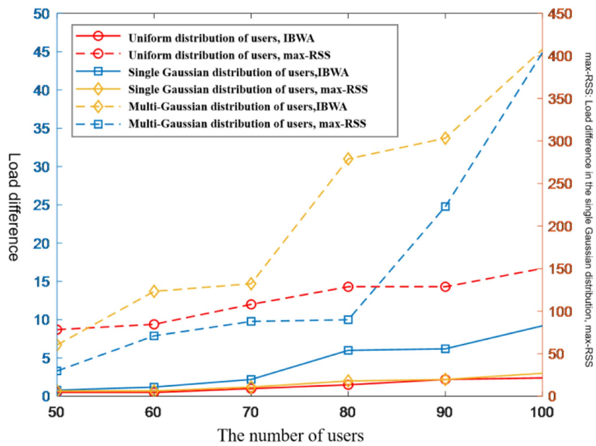

Fig. 3. Load difference among UAVs under IBWA and max-RSS algorithms.

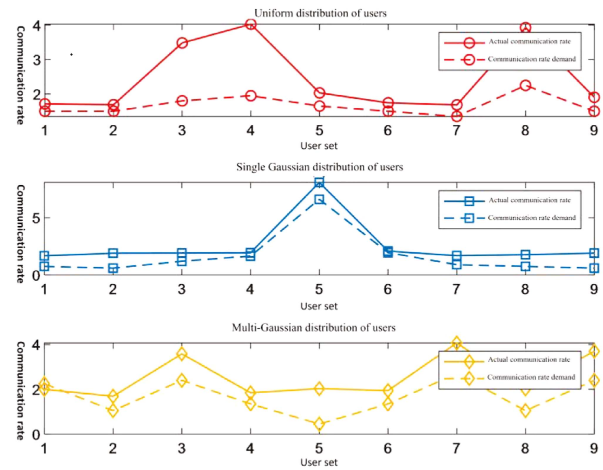

Fig. 4. The comparison between the communication rate demand and actual communication rate of each user set.

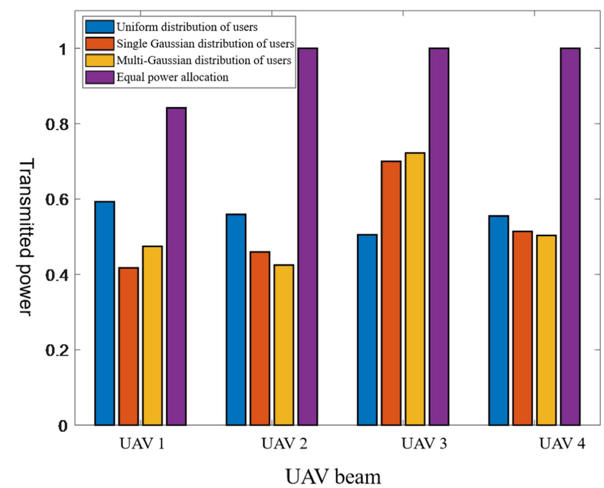

Fig. 5. Power consumption of UAVs under different user distributions.

The users on the ground are divided into 9 user sets according to the coverage of the beams. Unless otherwise specified, the number of UAVs is 4, and each UAV has three beams whose direction can be specified. Table I shows the general simulation parameters for the algorithms.

A. User Association

The effectiveness of the proposed IBWA algorithm using the users' communication rate demands as constraints and aimed at balancing load will be mainly verified in two aspects. The communication rate demand of a single user is 0.15 b/s/Hz. Simulation is performed according to the algorithm 1 to obtain simulation data.

Fig. 3 depicts the load difference among UAVs in different user distributions under the IBWA algorithm and the maximum received signal (max-RSS) based association mechanism, where the blue dashed line corresponds to the ordinate scale on the right. As can be seen from Fig. 3, comparing with the max-RRS mechanism, the proposed IBWA load balance association mechanism can significantly reduce the load difference among UAVs, and the load balance will be increasingly advantageous as the number of users increases.

Fig. 4 depicts the comparison between the actual communication rate and the communication rate demand of each user set in different user distributions. As can be seen, the solid line is always higher than the dashed line regardless of the fact whether the users conform to a uniform distribution, a single Gaussian distribution or a multi-Gaussian distribution, indicating that the communication rate demand of each user set is met. Accordingly, the effectiveness of the proposed algorithm is verified.

B. Power Allocation

The effectiveness of the power allocation algorithm will be mainly verified. The position of UAVs and the association between UAV beams and users are set according to the results of optimization in Section IV-A.

As can be seen from Fig. 5, by comparing with the equal allocation algorithm, using the proposed power algorithm, the total power consumption of UAVs is reduced regardless of the distribution of the users. The total power consumption of the UAV 3 in the multi-Gaussian distribution has the smallest reduction of about 0.3 W after optimization, while the total power consumption of the UAV 1 in the single Gaussian distribution has the largest reduction of about 0.6 W.

C. Position Deployment

This section focuses on the performance of the Algorithm 3, and the improved differential evolution algorithm will be compared with the particle swarm and genetic algorithms of the same type. As can be seen from Fig. 6, although the improved differential evolution algorithm basically converges to the same value as other two algorithms, the convergence is realized by fewer iterations.

D. The Number of UAVs

The effectiveness of the optimization of the number of UAVs will be mainly verified. The initial number of UAVs is 5, and their positions are randomly distributed. After the joint optimization,

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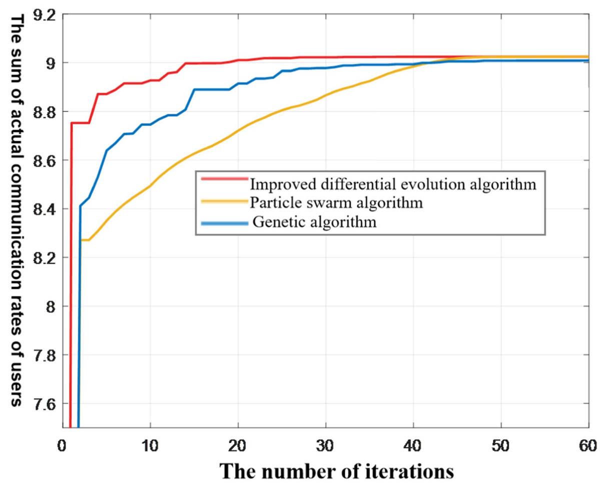

Fig. 6. The change of the sum of actual communication rates of users with the number of iterations under different algorithms.

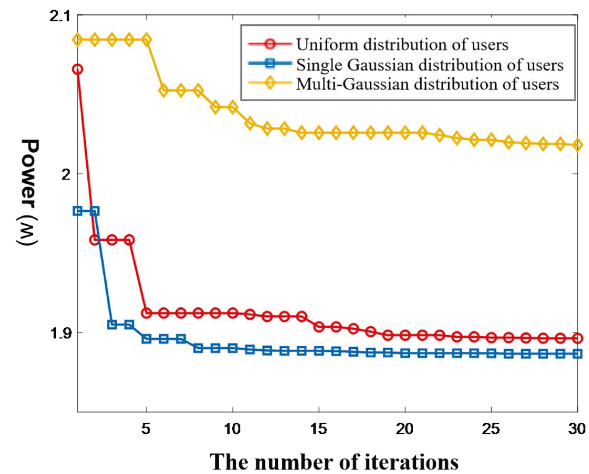

Fig. 8. The change of the total network power consumption with the number of iterations.

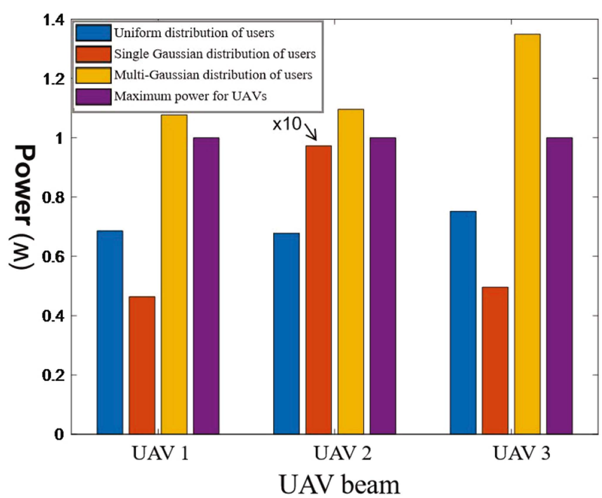

Fig. 7. Power consumption of UAVs under different distributions E. Total network power consumption.

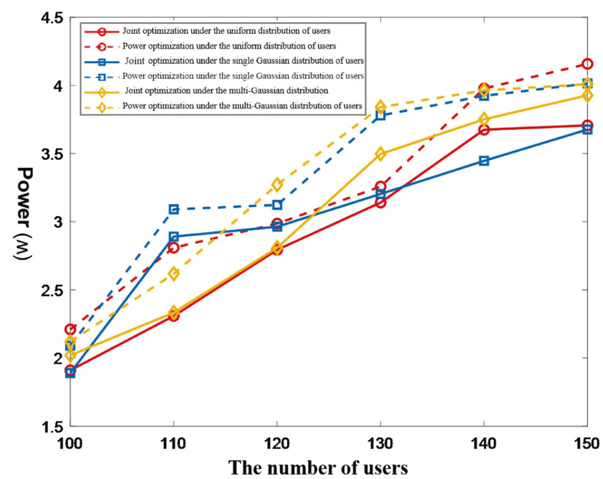

Fig. 9. The change of the total network power consumption with the number of users.

when the users are uniformly distributed, to meet the users' communication rate demands, the number of UAVs is at least 3; however, the number of UAVs is 4 when the users are in the single Gaussian distribution and the multi-Gaussian distribution. To further verify the correctness of the number of UAVs, the power consumption required to meet the users' communication rate demands is counted when the number of UAVs is 3. Due to the large power consumption value of UAV 2 when the user has a single Gaussian distribution, in order to facilitate display, the value is reduced by dividing it by 10.

As can be seen from Fig. 7, when the number of UAVs is 3, to meet the users' communication demands in the multi-Gaussian distribution or the single Gaussian distribution, the power consumption of UAVs exceeds the maximum value of 1 W. Particularly, when the users are in the single Gaussian distribution, the power required by the UAV 2 is close to 10 W, which is far greater than the maximum power value. Therefore, to meet the users' communication rate demands in the single Gaussian distribution and the multi-Gaussian distribution, the number of UAVs is at least 4.

The effectiveness of reducing the network power consumption by the inner layer alternate optimization algorithm will be verified. Fig. 8 depicts the change of the total network power consumption of UAVs after the joint optimization with the number of iterations when the number of UAVs is optimal. For unity, the number of UAVs is 4. As can be seen, after the joint optimization, the total network power consumption decreases with the increase of the number of iterations and finally converges. In the uniform distribution, single Gaussian distribution and multi-Gaussian distribution of the users, the total power consumption decreases from original 4 W to 1.9 W, 1.8 W and 2 W, respectively.

To further verify the necessity of the joint optimization, the joint optimization is compared with the situation where the power optimization is performed alone. As can be seen from Fig. 9, regardless of the distribution of the users, the total network power consumption will increase with the increase of the number of users, but the solid line is lower than the dashed line all the time. It indicates that the joint optimization can reduce the total network power consumption to a greater extent, compared with the situation where the power optimization is performed alone.

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V. Conclusion

In terms of energy efficient cooperative UAVs communications with multi-beams, this paper jointly optimize the association between UAV beams and user sets, the power consumption, position and number of UAVs. A nested multi-objective mixed integer non-convex optimization model is established and an alternate optimization scheme decoupling this optimization problem to inner and outer layer optimization is proposed. The inner layer optimization alternately optimizes three sub-problems, where the association problem between UAV beams and user sets is solved by the improved whale algorithm, the power allocation problem is solved by the successive convex approximation and the position optimization problem is solved by the improved differential evolution algorithm. The optimal or minimum number of UAVs is obtained by the outer layer optimization according to the results of inner layer alternate optimization. Numerical results show that: the proposed cooperative UAVs communications with multi-beam association and joint optimization scheme can flexibly adapt to the distribution of complex traffic tasks, reduce the power consumption and achieve the load balance among UAVs. In the future, an array antenna shaping vector determination algorithm that integrates the position information, channel information and association variables of UAVs and grids will be further studied.

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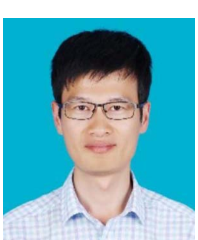

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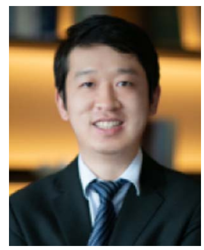

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